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0.1 Probability Spaces

Definition 0.1.1 ► Probability Spaces

A **probability space** is a measure space $(\Omega, \mathcal{F}, P)$ with

$$P : \mathcal{F} \rightarrow [0, 1]$$

Example 0.1.2 ► Discrete probability spaces

Let Ω be a countable set . Let $\mathcal{F}$ be the collection of all subsets of Ω . Let

$$P(A) = \sum_{\omega \in A} p(\omega), \quad \text{where } p(\omega) \geq 0 \text{ and } \sum_{\omega \in \Omega} p(\omega) = 1$$

Example 0.1.3 ► Measures on the real line.

Measures on $(\mathbf{R}, \mathcal{R})$ are defined by giving probability a **Stieltjes measure function** with the following properties:

- (i) F is nondecreasing.
- (ii) F is right continuous, that is, $\lim_{y \downarrow x} F(y) = F(x)$.

Theorem 0.1.4

Associated with each Stieltjes measure function F there is a unique measure μ on $(\mathbf{R}, \mathcal{R})$ with $\mu((a, b]) = F(b) - F(a)$

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When $F(x) = x$ the resulting measure is called **Lebesgue measure**

Definition 0.1.5 ▶ Semialgebra

A collection $\mathcal{S}$ of sets is said to be a **semialgebra** if (i) it is closed under intersection, that is, $S, T \in \mathcal{S}$ implies $S \cap T \in \mathcal{S}$, and (ii) if $S \in \mathcal{S}$ then S^c is a finite disjoint union of sets in $\mathcal{S}$.

Definition 0.1.6 ▶ Algebra

A collection $\mathcal{A}$ of subsets of Ω is called an **algebra** (or **field**) if $A, B \in \mathcal{A}$ implies A^c and $A \cup B$ are in $\mathcal{A}$.

Remark.

Since $A \cap B = (A^c \cup B^c)^c$, it follows that $A \cap B \in \mathcal{A}$. Obviously a σ -algebra is an algebra.

Lemma 0.1.7

If $\mathcal{S}$ is a semialgebra, then $\bar{\mathcal{S}} = \{\text{finite disjoint unions of sets in } \mathcal{S}\}$ is an algebra, called the **algebra generated by $\mathcal{S}$** .